

Assignment 2: Applying Microservices and Event-Driven Architecture in E-Commerce:

Scenario

Imagine developing an e-commerce platform with the following architecture:

- **Microservices Architecture:**
 - **Services:** User Management, Product Catalog, Shopping Cart, Order Processing, Payment, and Notification.
 - **Benefits:** Each service can be developed, deployed, and scaled independently, enhancing flexibility and resilience.
- **Event-Driven Architecture:**
 - **Mechanism:** Services communicate through events (e.g., "Order Placed", "PaymentProcessed").
 - **Benefits:** Promotes loose coupling and allows services to react to events asynchronously, improving scalability and responsiveness.

Applying SOLID Principles

- **Single Responsibility Principle:** Each microservice handles a specific business function.
- **Open/Closed Principle:** Services can be extended with new features without modifying existing code.
- **Liskov Substitution Principle:** Interchangeable service implementations without affecting the system's behavior.
- **Interface Segregation Principle:** Services expose only necessary interfaces, preventing clients from depending on unused functionalities.
- **Dependency Inversion Principle:** High-level modules depend on abstractions, allowing for flexible and testable code.

Observing DRY and KISS Principles

- **DRY (Don't Repeat Yourself):**
 - Utilize shared libraries for common functionalities like logging and authentication.
 - Centralize configuration management to avoid redundancy.
- **KISS (Keep It Simple, Stupid):**

- Design services with clear and concise responsibilities.
- Avoid over-engineering; implement features as needed.